KARUNGA JUNIOR SCHOOL
GRADE 9 INTEGRATED SCIENCE
END OF TERM 1 EXAMINATION

Teacher-prepared assessment resource | Kenya CBC

Learner's Name:	_____	Grade:	9
Date:	_____	Time:	2 hours 30 minutes

INSTRUCTIONS

1. Answer ALL questions.

2. Read each question carefully and show working where calculations are required.

3. Draw clear, labelled diagrams where requested.

4. Observe all laboratory safety requirements.

5. The examination carries 100 marks.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Structure of the Atom (8 marks)

(a) Name the three main subatomic particles. (3 marks)

(b) State the relative charge of a proton, neutron and electron. (3 marks)

(c) An atom has atomic number 12 and mass number 24. Determine the number of protons, electrons and neutrons in a neutral atom. (2 marks)

2. Metals, Alloys and Water (8 marks)

(a) State two physical properties commonly shown by metals. (2 marks)

(b) What is an alloy? (2 marks)

(c) Give one example of an alloy and one use of it. (2 marks)

(d) Distinguish between temporary and permanent hardness of water. (2 marks)

3. Nutrition in Plants (8 marks)

(a) Name the raw materials required for photosynthesis. (2 marks)

(b) State the role of chlorophyll in photosynthesis. (2 marks)

(c) Name the gas released during photosynthesis. (1 mark)

(d) State three factors that can affect the rate of photosynthesis. (3 marks)

4. Nutrition in Animals (8 marks)

(a) Define digestion. (2 marks)

(b) Name the organ where most absorption of digested food takes place in humans. (1 mark)

(c) State two functions of the small intestine that support digestion and absorption. (2 marks)

(d) Name one digestive enzyme and state the type of food it acts on. (2 marks)

(e) State one reason why a balanced diet is important. (1 mark)

5. Reproduction and Interdependence (8 marks)

(a) State two functions of a flower in a flowering plant. (2 marks)

(b) Distinguish between pollination and fertilisation. (2 marks)

(c) Define a food chain. (2 marks)

(d) State two ways human activities can disrupt an ecosystem. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

6. Atomic Structure and Isotopes (12 marks)

(a) An atom of element X is represented as $^{23}_{11}X$. State its: (i) atomic number; (ii) mass number; (iii) number of protons; (iv) number of neutrons. (4 marks)

(b) Explain why a neutral atom has no overall electrical charge. (2 marks)

(c) Element Y has two isotopes, Y-35 and Y-37. Explain what the term isotope means. (2 marks)

(d) State two ways in which models or symbols help scientists communicate information about atoms. (4 marks)

7. Metals, Alloys and Water Hardness (12 marks)

(a) Explain why copper is suitable for making electrical wires. Give two reasons. (4 marks)

(b) State two advantages of making alloys instead of using pure metals. (2 marks)

(c) Describe one method of removing temporary hardness of water and explain how it works. (3 marks)

(d) Explain why permanent hardness is not removed by boiling. (3 marks)

8. Photosynthesis and Plant Nutrition (12 marks)

(a) Write a word equation for photosynthesis. (2 marks)

(b) Describe an experiment that can be used to test a leaf for starch after exposure to sunlight. Include two safety precautions. (6 marks)

(c) Explain why a plant kept in darkness for a long time may have reduced starch in its leaves. (2 marks)

(d) State two ways plants use products of photosynthesis. (2 marks)

9. Human Nutrition and Digestion (12 marks)

(a) Describe the role of the mouth, stomach and small intestine in digestion. (6 marks)

(b) Explain the difference between mechanical digestion and chemical digestion. (2 marks)

(c) A learner regularly eats meals containing mainly starch but very little protein, vitamins and minerals. Explain two possible effects of this unbalanced diet. (4 marks)

10. Reproduction and Interdependence of Life (12 marks)

(a) Describe the sequence of events from pollination to seed formation in a flowering plant. (5 marks)

(b) Explain two adaptations of seeds or fruits that help in dispersal. (4 marks)

(c) Study this food chain: grass → grasshopper → frog → snake. Identify: (i) the producer; (ii) the primary consumer; (iii) the secondary consumer. (3 marks)

MARKING SCHEME
GRADE 9 INTEGRATED SCIENCE END OF TERM 1

TOTAL: 100 MARKS

1. Structure of the Atom (8 marks)

(a) Proton, neutron and electron. (3)

(b) Proton +1; neutron 0; electron −1. (3)

(c) Protons = 12; electrons = 12; neutrons = 24 − 12 = 12. (2)

2. Metals, Alloys and Water (8 marks)

(a) Any two: lustrous, malleable, ductile, good conductors of heat/electricity, generally strong. (2)

(b) A mixture of a metal with one or more other elements, at least one of which is a metal, made to improve properties. (2)

(c) Accept brass—musical instruments/taps; bronze—statues/medals; steel—buildings/tools; stainless steel—cutlery. One valid example and use. (2)

(d) Temporary hardness can be removed by boiling; permanent hardness is not removed by boiling. (2)

3. Nutrition in Plants (8 marks)

(a) Carbon dioxide and water. (2)

(b) Absorbs/traps light energy for photosynthesis. (2)

(c) Oxygen. (1)

(d) Any three: light intensity, carbon dioxide concentration, temperature, water availability or chlorophyll amount. (3)

4. Nutrition in Animals (8 marks)

(a) Breakdown of large, insoluble food molecules into small, soluble molecules that can be absorbed. (2)

(b) Small intestine. (1)

(c) Any two: completes digestion; has villi for large surface area; absorbs nutrients into blood/lymph; receives digestive juices. (2)

(d) E.g. amylase—starch; protease—proteins; lipase—fats. (2)

(e) Any valid reason: growth, repair, energy, immunity or healthy body function. (1)

5. Reproduction and Interdependence (8 marks)

(a) Any two: produces male/female gametes; attracts pollinators; enables pollination and fertilisation; develops seeds/fruits after fertilisation. (2)

(b) Pollination is transfer of pollen from anther to stigma; fertilisation is fusion of male and female gamete nuclei. (2)

(c) A sequence showing transfer of food/energy as one organism is eaten by another. (2)

(d) Any two: deforestation, pollution, overgrazing, overfishing, habitat destruction, use of harmful chemicals or introduction of invasive species. (2)

6. Atomic Structure and Isotopes (12 marks)

(a) Atomic number 11; mass number 23; protons 11; neutrons 23 − 11 = 12. (4)

(b) It has equal numbers of positively charged protons and negatively charged electrons, so the charges cancel. (2)

(c) Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons/mass numbers. (2)

(d) Any two explained uses, up to 4: show arrangement of subatomic particles; communicate atomic/mass numbers; compare elements/isotopes; provide simplified representations for scientific explanation.

7. Metals, Alloys and Water Hardness (12 marks)

(a) Copper is a good electrical conductor and is ductile/easily drawn into wires; it is also reasonably durable and corrosion-resistant. Award 2 per explained reason. (4)

(b) Any two: greater strength/hardness; improved corrosion resistance; improved durability; improved suitability for a particular use. (2)

(c) Boil the water; dissolved calcium/magnesium hydrogencarbonates decompose to form insoluble precipitates which can be removed by filtration/settling. (3)

(d) Permanent hardness is commonly caused by dissolved calcium/magnesium sulphates or chlorides, which do not decompose into insoluble precipitates on boiling. (3)

8. Photosynthesis and Plant Nutrition (12 marks)

(a) Carbon dioxide + water → glucose + oxygen (in the presence of light and chlorophyll). (2)

(b) Any correct sequence: boil leaf in water; heat in ethanol using a hot-water bath to remove chlorophyll; rinse/soften in warm water; place on white tile; add iodine solution; blue-black colour indicates starch. Award up to 4 for method and 2 for safety (ethanol is flammable—use water bath/no naked flame; handle hot apparatus with forceps). (6)

(c) Without light, photosynthesis is reduced or stops; stored starch is used in respiration and is not replenished adequately. (2)

(d) Any two: respiration after conversion to glucose; stored as starch; used to make cellulose, oils or proteins (with mineral nutrients); growth and repair. (2)

9. Human Nutrition and Digestion (12 marks)

(a) Mouth: chewing breaks food; saliva moistens food and amylase begins starch digestion. Stomach: churns food; acid provides suitable conditions and protease begins protein digestion. Small intestine: receives bile/enzymes, completes digestion and absorbs nutrients through villi. Award up to 2 per organ. (6)

(b) Mechanical digestion physically breaks food into smaller pieces without changing its chemical nature; chemical digestion uses enzymes/chemical substances to break large molecules into smaller soluble molecules. (2)

(c) Any two explained effects: poor growth/repair due to protein deficiency; deficiency diseases due to lack of vitamins/minerals; low immunity; tiredness or poor health. (4)

10. Reproduction and Interdependence of Life (12 marks)

(a) Pollen lands on stigma; pollen grain germinates; pollen tube grows down the style; male nucleus travels to ovule; male and female nuclei fuse; ovule develops into seed and ovary develops into fruit. Award any five correct linked stages. (5)

(b) Any two explained: hooks/barbs attach to animals; wings/hairs enable wind dispersal; fleshy fruits attract animals which carry or excrete seeds; floating fruits/seeds disperse by water; explosive pods scatter seeds. (4)

(c) Producer—grass; primary consumer—grasshopper; secondary consumer—frog. (3)

GENERAL MARKING GUIDANCE

Award method marks for scientifically sound explanations and procedures even where wording differs from the guide.

Accept other accurate examples and explanations relevant to the question.

For practical questions, award marks for correct sequence, observation, interpretation and safety precautions as indicated.

Teachers should confirm that the assessed sub-strands have been taught. This paper is a teacher-prepared resource based on the Term 1 sequence in the school's Grade 9 Integrated Science scheme; it is not an official KICD examination.

END OF EXAMINATION